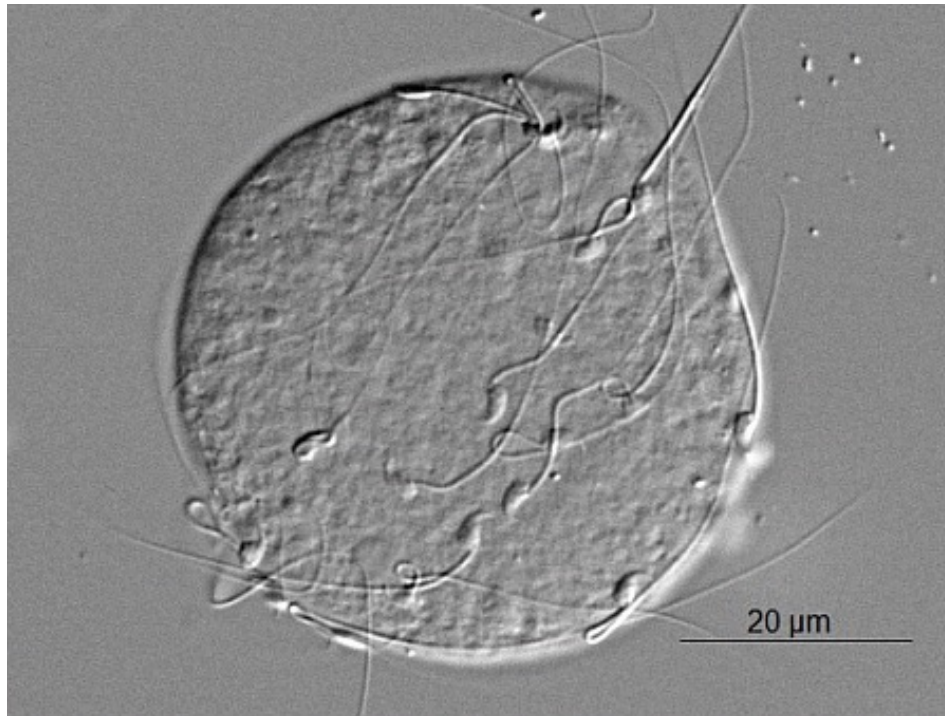


Non-Mendelian Genetics II (Cytoplasmic inheritance / Maternal effect)

Readings: Ch3 p97-102 [section 3.5]

- Cytoplasmic inheritance
- Organelle transmission in animals
- Mitochondria dysfunction and diseases

We are the product of a sperm and oocyte joining to form a diploid cell...



<https://doi.org/10.3390/ijms150610652>

- offspring inherit many things from their mothers through the cytoplasm of the oocyte

- proteins
- mRNAs (and other RNAs)
- organelles

- non-Mendelian inheritance

“cytoplasmic inheritance”

or

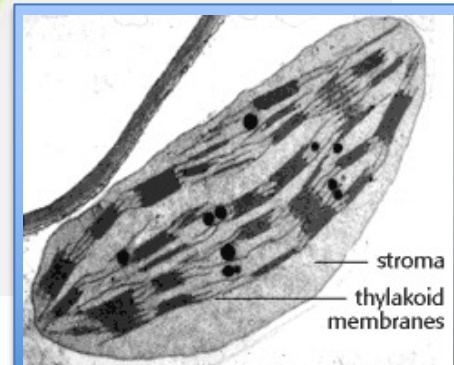
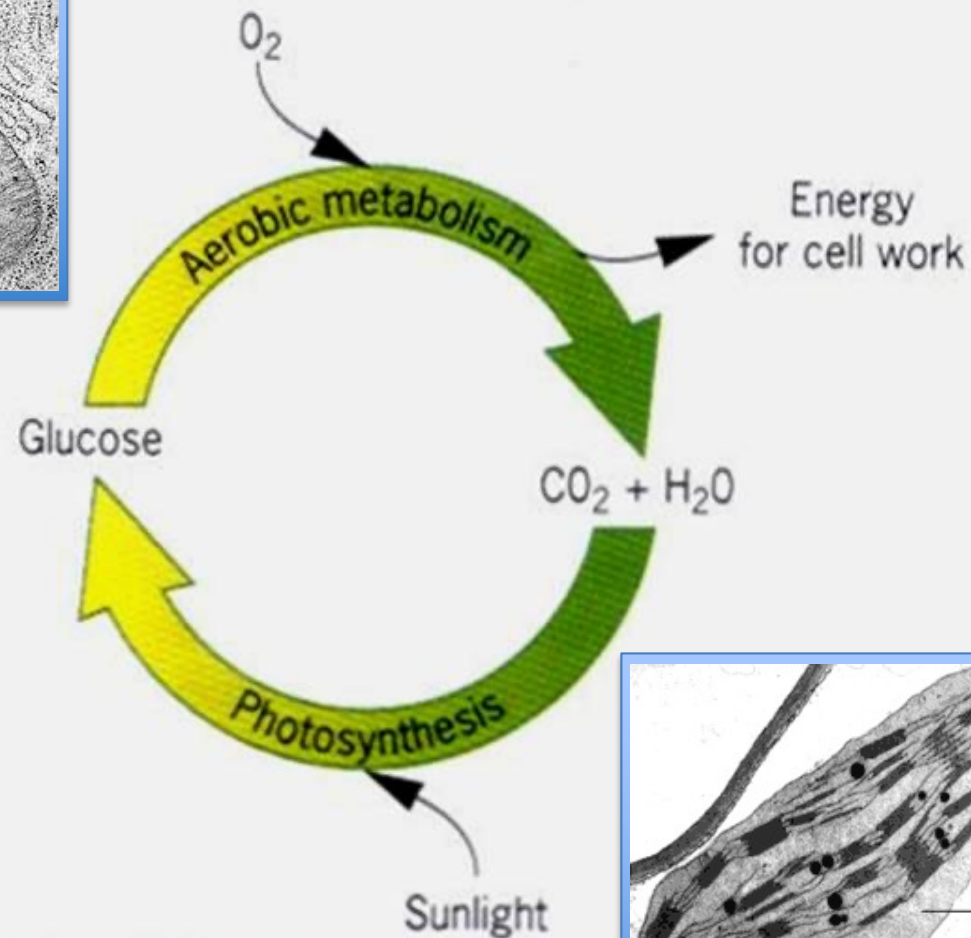
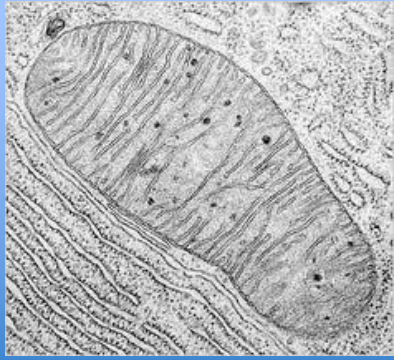
“maternal inheritance”

or

“extranuclear inheritance”

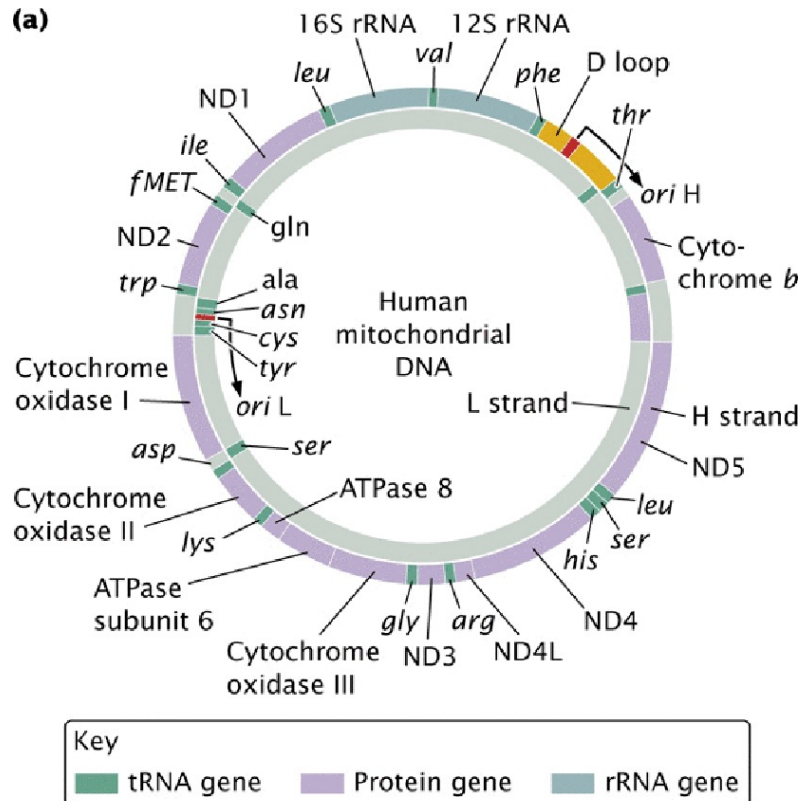
Cytoplasmic inheritance: Organelle genetics

Mitochondria

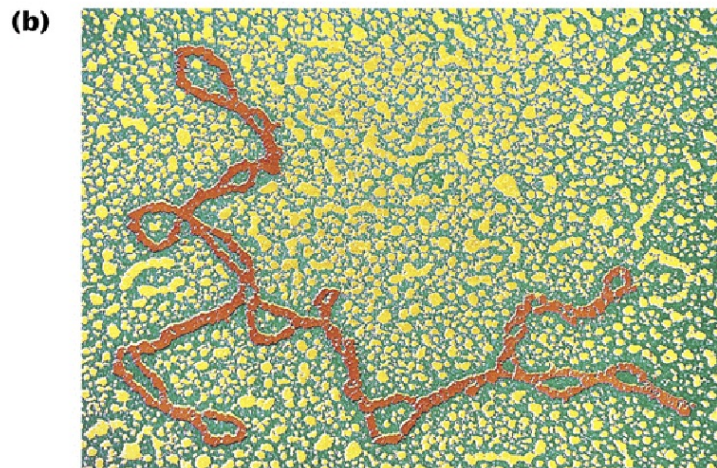


Chloroplast

Mitochondrial genome in humans



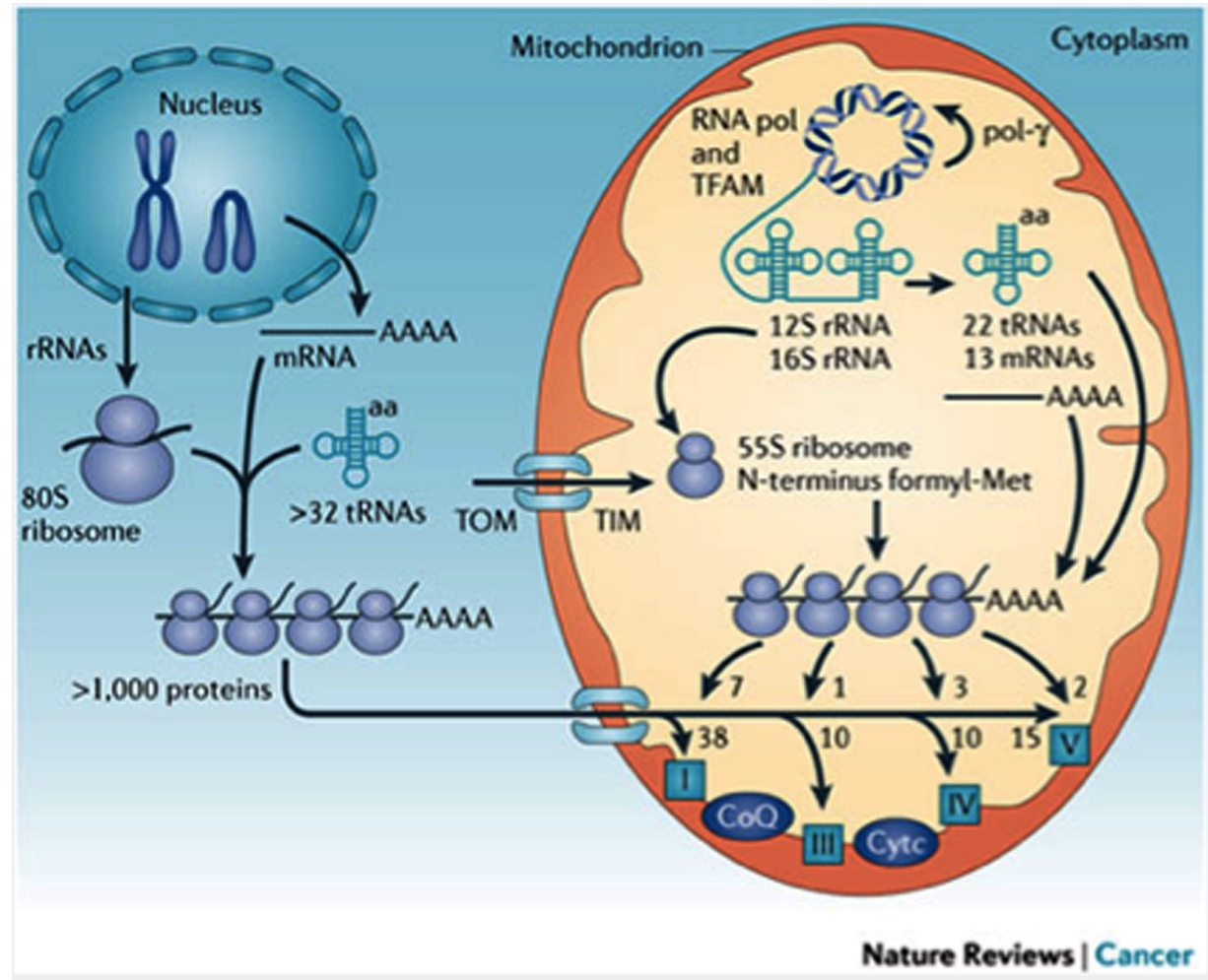
- 16,569 nt, circular DNA molecule
- dozens to hundreds of mitochondria/cell, each with multiple mtDNA copies - 1% of cellular DNA!
- encodes 13 proteins and 24 rRNA/tRNAs
- Gene sequences most similar to α -proteobacteria – **endosymbiotic theory**



BUT... over 1000 proteins in a mt... HOW?

Nuclear genes contribute the bulk of mitochondrial proteins

- 99% of mt proteins are encoded in the nuclear genome
- The 13 mtDNA-encoded proteins all involved in electron transport chain
- Other proteins produced in cytoplasm, imported
- Nuclear mt genes are most similar to bacterial genes – must have moved to nucleus over time



Wallace DC. Nature Rev Cancer (2012)

Factors influencing replication of mitochondrial genome

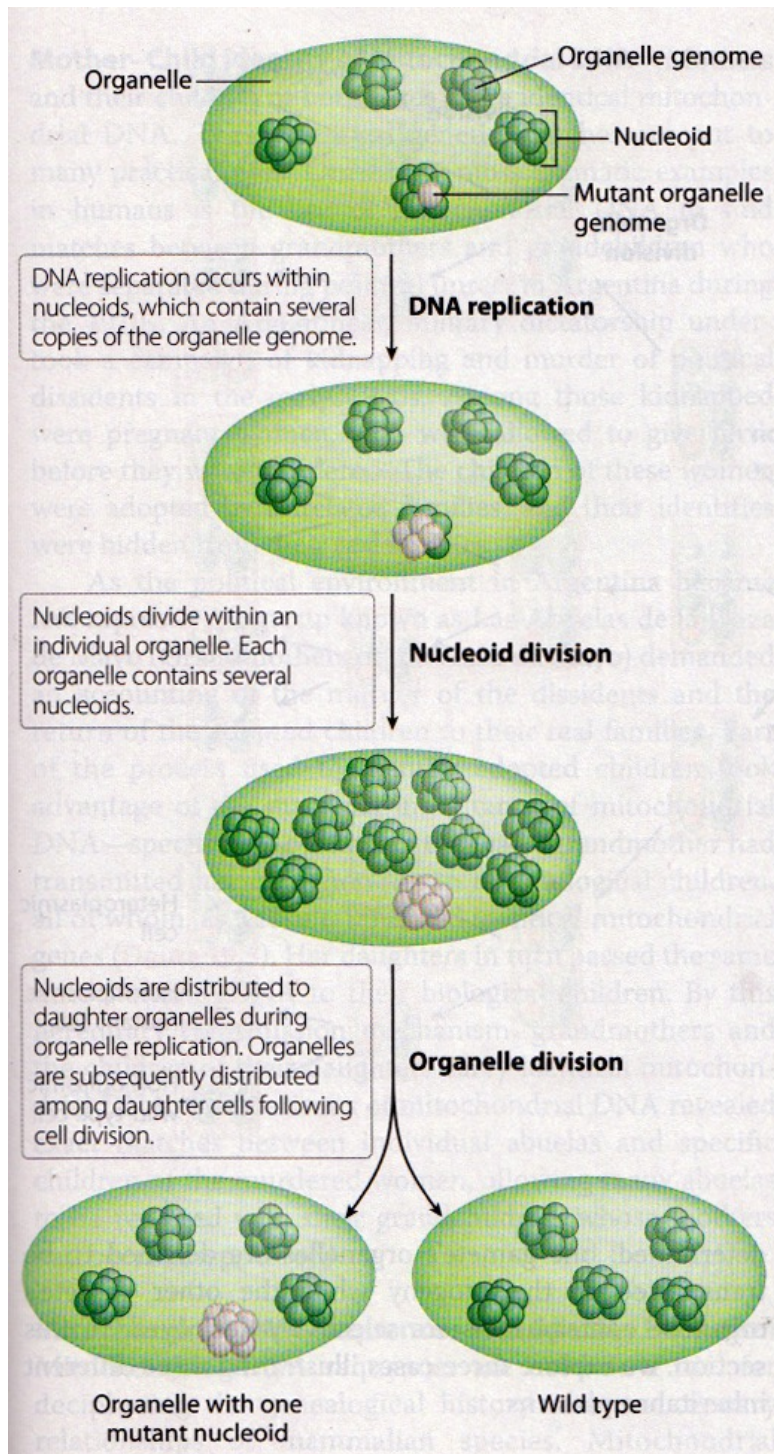
1. **Mitochondrial DNA is replicated within nucleoids** (protein/DNA particles in which mtDNA is packaged) – several mt genomes/nucleoid

Independent of cell cycle/nuclear DNA replication

2. The **nucleoids can divide within an organelle**

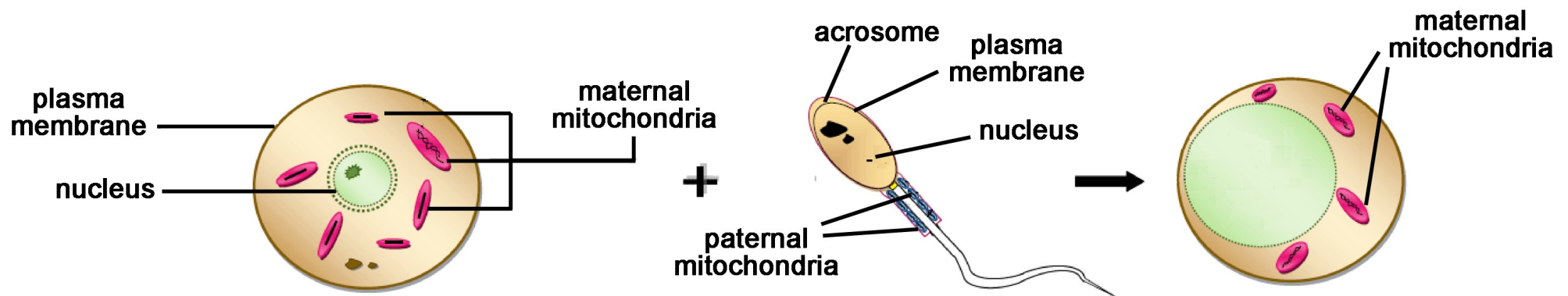
3. **Mitochondria themselves divide**

arise by division of existing ones (not made *de novo*)



Mitochondria are inherited from the mother in animals – cytoplasmic inheritance

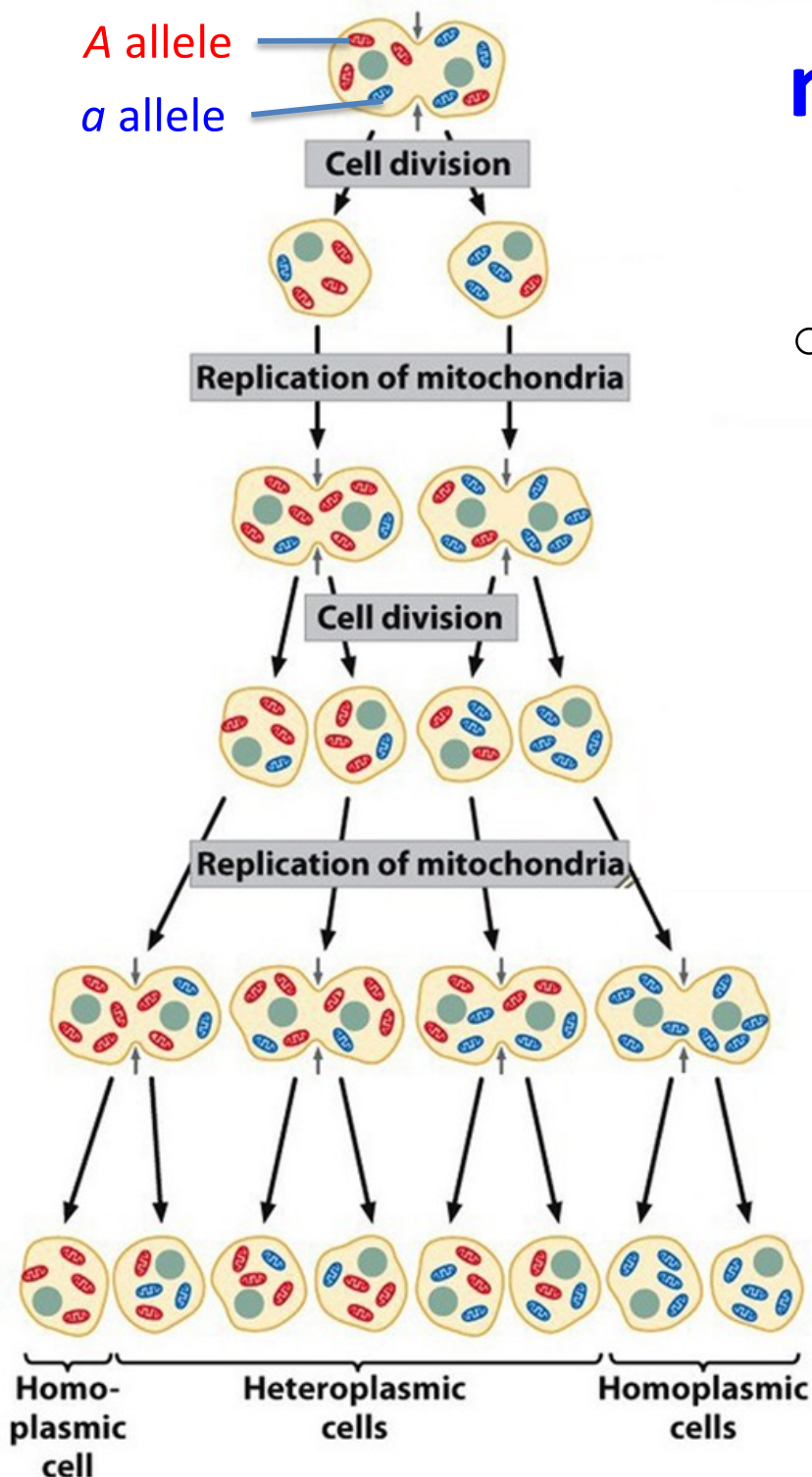
- oocyte and sperm both contain mitochondria, but there are far more in oocyte (>100,000 vs ~100)



- Upon fertilization, the few sperm mitochondria are destroyed (or consumed) by the oocyte

**All mitochondria inherited from the mother
(maternal inheritance)**

(mtDNA can be used as a tool to trace maternal heritage)



mtDNA mutations can lead to heteroplasmy

- mt genes ~1000-fold higher rate of mutation than nuclear genes
more frequent DNA replication
no DNA repair
- Spontaneous mtDNA mutations can lead to two distinct mt populations within a single cell - **heteroplasmy**
- Random segregation of organelles at mitosis (or meiosis) can lead to cells **homoplasmic** for mtDNA mutation

Pierce Fig 21.4

Most mitochondrial proteins are encoded by nuclear genes. Which of the following statements about the inheritance of these genes is true?

- A. they will be inherited in a non-Mendelian manner
- B. only the mother's genotype will matter for the phenotype of the offspring
- C. the mother's alleles will act in a dominant manner
- D. a wild-type paternal allele will most likely complement a defective maternal allele

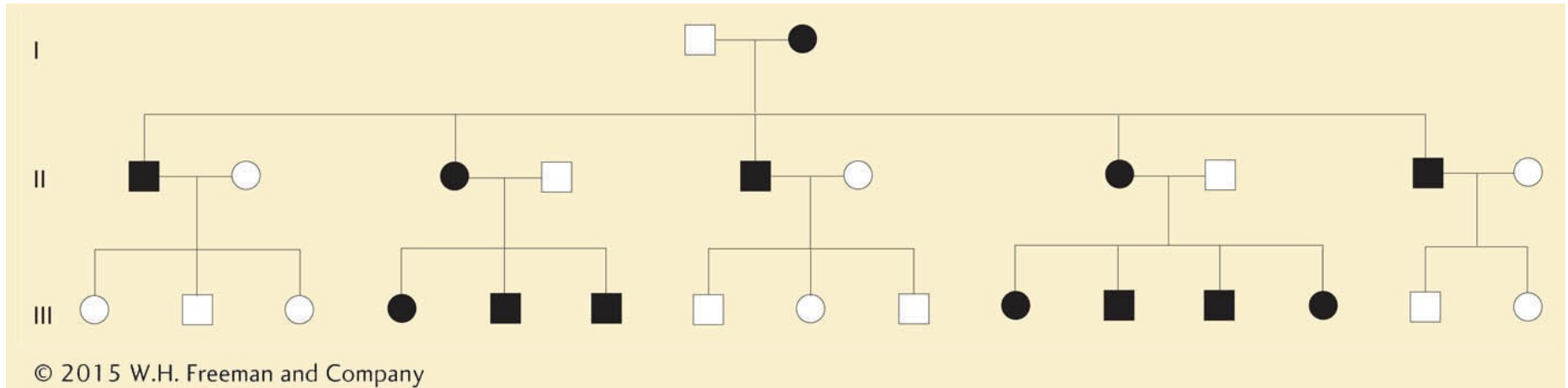
Table. Recognizable syndromes of mitochondrial dysfunction.

Syndrome and features	Genetics
Leigh syndrome Neonatal subacute encephalopathy with bilateral symmetric midbrain and basal ganglia necrosis on MRI	Autosomal recessive, mitochondrial DNA, X-linked
Pearson syndrome Sideroblastic anemia, pancytopenia, exocrine pancreatic insufficiency, and renal tubulopathy	Mitochondrial DNA
MERRF Myoclonic epilepsy with ragged-red fibres on muscle biopsy	Mitochondrial DNA
NARP Neurogenic weakness, ataxia, and retinitis pigmentosa	Mitochondrial DNA
MELAS Mitochondrial encephalopathy with lactic acidosis and stroke-like episodes	Mitochondrial DNA
Alpers syndrome Encephalopathy, seizures, and hepatic dysfunction	Autosomal recessive, autosomal dominant
MNGIE Mitochondrial neurogastronintestinal encephalopathy	Autosomal recessive
Kearns-Sayre syndrome External ophthalmoplegia, pigmentary retinopathy, elevated CSF protein, cerebellar ataxia, and cardiac conduction defects	Mitochondrial DNA; often sporadic
MIDD Maternally inherited diabetes and deafness	Mitochondrial DNA
SANDO Sensory ataxia, neuropathy, dysarthria, and ophthalmoplegia	Autosomal dominant

Some human diseases are caused by dysfunction in mitochondria

- Affect at least 1/5000 people
- Progressive
- Multi-system disease (high energy-demand tissues)

Pedigree of human mitochondrial disease caused by mitochondrial genes



- Children of an affected mother **all have the disease** (hence, maternal inheritance)
- Children of an affected father **never have the disease**.
- ***In some cases where a mother has a mixture of affected and unaffected mitochondria (**heteroplasmic**), her children **may be normal**.***

The beginning of the end for mitochondrial disease?

nature

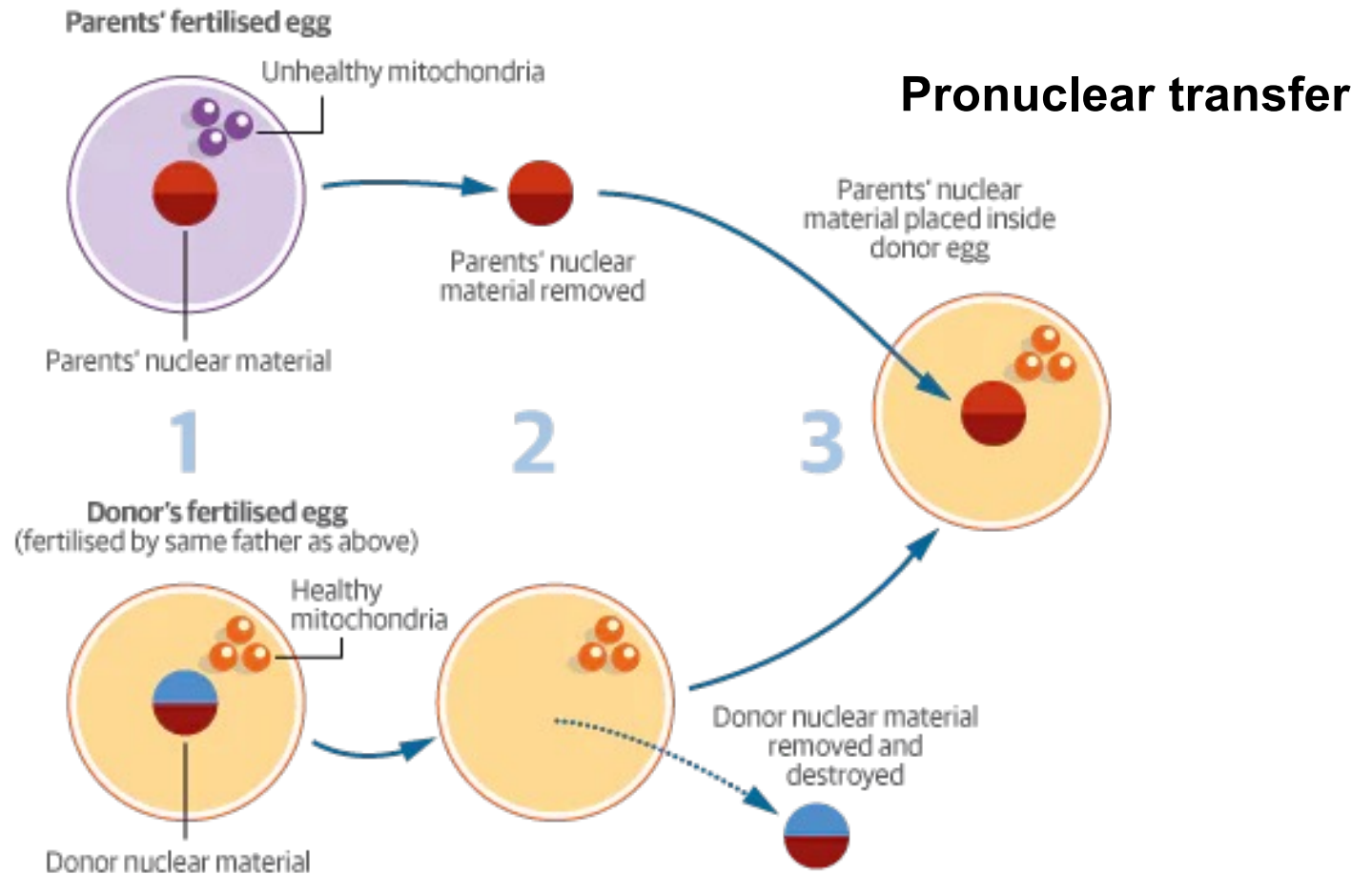
Vol 465 | 6 May 2010 | doi:10.1038/nature08958

LETTERS

Pronuclear transfer in human embryos to prevent transmission of mitochondrial DNA disease

Lyndsey Craven¹, Helen A. Tuppen¹, Gareth D. Greggains^{3,4}, Stephen J. Harbottle³, Julie L. Murphy¹, Lynsey M. Cree¹, Alison P. Murdoch^{3,5}, Patrick F. Chinnery¹, Robert W. Taylor¹, Robert N. Lightowlers¹, Mary Herbert^{3,4,5} & Douglass M. Turnbull^{1,2,5}

The beginning of the end for mitochondrial disease?



- Three-person In vitro fertilization: resulting embryo has three parents